

ALETHEIA

Aletheia Fusion Intelligence — AFI

DETECTOR PACK EXPANSION

Complete Pack Library & Long-Horizon Expansion Map

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The Core Thesis

Everything is signals. Every digital system — biological sensor networks, financial rails, enterprise identity layers, manufacturing floors, power grids, robotic actuators, AI agents — produces behavioral signals as a byproduct of existing. The question is who learns to read them intelligently.

AFI's fusion engine doesn't detect cyber attacks specifically. It detects behavioral anomalies in signal streams. The connector library determines which signals enter the engine. The detector packs determine which behavioral geometries the engine recognises. The fusion math underneath is domain-agnostic by architecture — not by analogy.

The fusion engine does not care what signals it fuses. It cares about behavioral geometry — the shape of how things move, the timing, sequence, and structural signature of normal versus anomalous. This is not a metaphor. It is literally how the codebase is architected.

Master Pack Library — Complete Overview

Current confirmed packs, build sequence, and long-horizon expansion targets. Total detector count target: 100+ across all packs.

Pack	Ship Phase	Detectors	Priority	Regulatory Angle
MSP Operations Pack	NOW — Ships First	6	Critical	MSP operational security; cross-tenant credential isolation
Healthcare Pack (HIPAA)	Phase 1	8	High	HIPAA §164.312, PHI access audit, breach notification

Pack	Ship Phase	Detectors	Priority	Regulatory Angle
Financial Pack	Phase 1	8	High	SOX, PCI-DSS, GLBA, wire fraud indicators
Cloud Security Pack	Phase 1–2	10	High	CSA CCM, AWS/Azure security frameworks, CSPM gap-fill
Identity Risk Pack	Phase 1–2	6	High	Zero Trust, NIST 800-207, privilege management audits
Legal Pack	Phase 2	6	Medium	Matter confidentiality, ABA duties, data handling compliance
Email & Collaboration Pack	Phase 2 (post-13)	8	Medium	Email DLP, M365 security, OAuth governance
Endpoint Pack	Phase 2 (post-13)	7	Medium	EDR augmentation, MITRE ATT&CK coverage mapping
Compliance Pack	Phase 2	6	Medium	SOC 2, ISO 27001, CMMC, audit chain integrity
OT / Industrial Pack	Phase 3	8	Medium	NERC CIP, IEC 62443, CISA ICS advisories
Insurance Underwriting Pack	Phase 3	7	Medium	Cyber insurance carrier requirements, risk scoring
Static Systems / Physical Security Pack	Phase 3	6	Medium	Physical-digital convergence, door/badge/camera anomaly
Supply Chain Pack	Phase 3–4	7	Medium	NIST SP 800-161, vendor risk management, SBOM integrity
Telecommunications Pack	Phase 4	7	Medium	FCC robocall rules, STIR/SHAKEN, fraud detection
AI Agent Security Pack	Phase 4	8	High	Emerging AI governance frameworks, agentic oversight
Robotics & Autonomous Systems Pack	Phase 4–5	7	Strategic	Emerging OT/robotics standards, autonomous system integrity
Defense & Intelligence Pack	Phase 5	8	Strategic	CMMC Level 3, NIST 800-171, DoD supply chain requirements

MSP Operations Pack

Ships first — protects the MSP's own business before protecting their clients

SHIP NOW

Every other pack serves the MSP's clients. The MSP Operations Pack serves the MSP directly. The first conversation isn't 'this will help your clients.' It's 'this will protect your own business.' Faster yes. Stronger trust signal.

Detector	Signal Type	Weight	Gate	Notes
Cross-Tenant Credential Usage	Identity / Auth	1.3	0.65	Admin credential active across multiple client tenants simultaneously
Admin Login Across Multiple Clients	Identity / Auth	1.2	0.60	Same identity authenticating to >3 client environments within short window
Shared Credential Patterns	Identity / Auth	1.2	none	Credential entropy patterns suggesting shared or reused MSP admin keys
Unusual RMM Command Propagation	Endpoint / RMM	1.3	0.70	RMM-issued commands spreading across client environments atypically
MSP Tool Privilege Escalation	Privilege	1.2	0.60	MSP management tooling accessing scopes beyond operational baseline
After-Hours Multi-Client Access	Behavioral	1.0	none	MSP operator accessing multiple client environments outside business hours

Why This Matters to Every MSP

MSPs are the single highest-value target in the supply chain. One compromised MSP credential reaches every client environment they manage. The 2021 Kaseya VSA attack pushed ransomware to over 1,500 downstream businesses through a single MSP toolchain. This pack makes AFI the first

security product that protects the MSP's own operational infrastructure, not just their clients. The opening pitch writes itself.

Healthcare Pack (HIPAA)

PHI access anomalies, breach precursors, and audit-ready detection for regulated clinical environments

**Phase 1 — High
Priority**

Detector	Signal Type	Weight	Gate	Notes
PHI Bulk Export Anomaly	Data Exfiltration	1.3	0.65	Volume of patient record exports exceeds clinical baseline significantly
After-Hours Patient Record Access	Access Anomaly	1.1	none	EHR access outside operator's normal hours — correlated with identity
Unusual EHR Query Patterns	Behavioral	1.1	none	Query scope, record type, or depth deviating from clinician baseline
Mass Patient Lookup Behavior	Fan-Out	1.2	0.60	Single identity accessing broad patient list without clinical justification pattern
Doctor Account Lateral Movement	Lateral Movement	1.2	0.65	Clinical identity accessing systems outside their specialty/department
Medical Device Access Anomaly	OT / Device	1.1	none	Unusual authentication or command patterns to networked medical devices
Insurance / Billing Data Access	Data Access	1.0	none	Clinical identity accessing financial or billing records atypically
Record Amendment Burst	Integrity	1.1	0.60	High-volume record modification events in compressed timeframe

HIPAA §164.312(b) requires audit controls — 'implement hardware, software, and/or procedural mechanisms that record and examine activity in information systems.' AFI

Sentinel Evidence Reports fulfill this requirement directly and become exhibit-ready for breach investigation or OCR audit.

Financial Pack

Wire fraud precursors, trading anomalies, and privilege drift in environments where a single incident has material P&L consequences

**Phase 1 — High
Priority**

Detector	Signal Type	Weight	Gate	Notes
Wire Transfer Precursor Pattern	Fraud / Exfil	1.4	0.70	Behavioral sequence preceding wire transfer: lookup → modify → initiate
After-Hours Transaction Access	Access Anomaly	1.2	0.60	Financial system access outside business hours correlated with identity
Trading System Privilege Escalation	Privilege	1.3	0.65	Privilege gradient into execution or settlement systems above baseline
Rapid Account Privilege Drift	Privilege	1.2	none	Accumulated permissions across financial systems exceeding role baseline
Payment Rail Access Anomaly	Access Anomaly	1.3	0.70	ACH/SWIFT/FedWire system access from identity without payment history
Bulk Beneficiary Record Modification	Integrity	1.3	0.65	High-volume changes to payment destination records in compressed window
Cross-System Financial Data Aggregation	Exfiltration	1.2	0.60	Single identity querying multiple financial databases unusually in sequence
Trading Halt / Override Attempt	Policy	1.4	0.75	Attempt to access override or emergency functions in trading platforms

Financial environments carry the highest cost-per-incident of any sector. A single undetected wire fraud event can exceed the entire annual cost of AFI deployment. Framing: AFI isn't a security cost in financial environments — it's a loss prevention investment with a calculable ROI.

Cloud Security Pack

IAM drift, cross-region anomalies, storage exfiltration, and misconfiguration-adjacent behavioral signals

**Phase 1–2 —
High Priority**

Cloud environments generate enormous volumes of API and IAM signals. Microsoft Sentinel covers this space with per-workspace billing that punishes MSPs at scale. AFI's structural behavioral approach costs the same regardless of event volume — and the fusion engine catches coordinated cloud attacks that per-rule SIEM approaches miss entirely.

Detector	Signal Type	Weight	Gate	Notes
Unusual Cloud Resource Provisioning	Infrastructure	1.2	none	Compute/storage resources provisioned outside approved regions or types
Cross-Region Access Anomaly	Access Anomaly	1.2	0.60	API calls originating from or targeting regions outside operational baseline
Storage Exfiltration Pattern	Data Exfiltration	1.3	0.65	Bulk object enumeration + download sequence in blob/S3/GCS stores
IAM Privilege Drift	Privilege	1.3	0.65	Accumulated IAM permissions exceeding role baseline across cloud services
API Key Misuse	Identity / Auth	1.2	0.60	API key activity pattern deviating from service's normal call profile
Service Account Lateral Movement	Lateral Movement	1.3	0.65	Service account accessing cloud services outside its functional scope
Cloud Console Access Anomaly	Access Anomaly	1.1	none	Management console access from unusual identity, IP, or time pattern
Secrets Manager Access Burst	Exfiltration	1.3	0.70	High-volume secrets retrieval — precursor to

Detector	Signal Type	Weight	Gate	Notes
				credential harvesting
Network Policy Override Attempt	Policy	1.3	0.70	Attempt to modify security groups, VPC rules, or firewall policies
Data Pipeline Redirection	Exfiltration	1.2	0.60	ETL/data pipeline destination change — data exfil via trusted pipeline

Identity Risk Pack

Deep identity behavioral analysis beyond the core Tier 1 detectors — for environments where identity is the primary attack surface

Phase 1–2

Detector	Signal Type	Weight	Gate	Notes
Rapid Identity Attribute Changes	Identity	1.2	none	MFA methods, email, phone, recovery options changed in compressed window
Cross-Identity Synchronization	Coordinated Attack	1.3	0.65	Multiple identities exhibiting correlated behavioral changes simultaneously
Privilege Accumulation Pattern	Privilege	1.2	0.60	Slow accumulation of permissions across systems over extended window
Dormant Identity Reactivation	Behavioral	1.1	0.60	Identity inactive >60 days suddenly accessing sensitive resources
Identity Impersonation Signal	Identity	1.3	0.70	Behavioral profile mismatch suggesting credential theft or delegation abuse
Service Account Humanization	Behavioral	1.2	0.60	Service account exhibiting human-like behavioral

Detector	Signal Type	Weight	Gate	Notes
				timing — possible takeover

Legal Pack

Matter confidentiality, pre-departure document hoarding, and cross-client data exposure in legal environments

Phase 2

Detector	Signal Type	Weight	Gate	Notes
Matter Confidentiality Breach	Data Access	1.3	0.65	Attorney accessing matters outside their assigned client portfolio
Pre-Departure Document Hoarding	Exfiltration	1.4	0.70	Bulk document download/export correlated with calendar/HR signals
Unusual Case File Access	Access Anomaly	1.2	none	Access to sealed, archived, or opposing party-adjacent matter files
Cross-Client Data Exposure	Data Access	1.3	0.65	Data flows between logically separated client matter environments
Privileged Document Download Burst	Exfiltration	1.2	0.60	High volume download of attorney-client privileged documents
After-Hours Matter Access	Access Anomaly	1.0	none	Matter access pattern anomaly outside attorney's historical hours

Email & Collaboration Pack

OAuth anomalies, bulk forwarding rules, and external exfiltration patterns across M365 and Google Workspace

Phase 2 (post-Phase 13)

Detector	Signal Type	Weight	Gate	Notes
Bulk Email Forwarding Rule Creation	Exfiltration	1.3	0.65	Auto-forward rules targeting external domains created post-compromise
Unusual External Sharing Pattern	Data Exfiltration	1.2	0.60	SharePoint/Drive share events to external domains above baseline
OAuth App Grant Anomaly	Identity / Auth	1.3	0.65	High-privilege OAuth app granted by non-admin in compressed window
External Domain Exfiltration Pattern	Data Exfiltration	1.3	0.65	Email or file transfer volume to specific external domain deviating from baseline
Inbox Rule Modification Burst	Persistence	1.2	0.60	Multiple inbox rule changes in short window — BEC persistence indicator
Calendar Invite to External Unknown	Reconnaissance	1.0	none	Mass calendar invites to unknown external recipients — social engineering signal
Teams/Slack Data Export Anomaly	Exfiltration	1.2	0.60	Collaboration platform export activity outside approved workflows
Email Delegation Grant to External	Identity	1.3	0.70	Mailbox delegation granted to external or unknown identity

Endpoint Pack

EDR signal fusion — AFI sits above endpoint tools and fuses their signals with network, identity, and cloud context

Phase 2 (post-Phase 13)

AFI doesn't replace EDR. It fuses EDR signals with everything else. A process injection alert from CrowdStrike becomes a confirmed incident when the fusion engine correlates it with an unusual network connection, a first-seen external IP, and an identity that authenticated 6 minutes ago from a new device. That correlation is the incident. The EDR alert alone is noise.

Detector	Signal Type	Weight	Gate	Notes
Process Injection Pattern	Endpoint	1.3	0.65	Memory injection techniques detected by EDR — fused with network context
Persistence Mechanism Detection	Persistence	1.3	0.65	Registry, scheduled task, or service creation outside baseline
Lateral Tool Transfer	Lateral Movement	1.2	0.60	Tool or script moved across systems — fused with identity activity
EDR Alert Burst Correlation	Meta-Signal	1.2	0.60	Multiple EDR alerts across identities/systems in correlated timeframe
Living-off-the-Land Behavioral Pattern	Evasion	1.3	0.65	Legitimate tools (PSEXEC, WMI, certutil) used in atypical sequences
Unusual Child Process Sequence	Endpoint	1.1	none	Parent-child process relationship deviating from software baseline
Credential Dumping Precursor	Credential Theft	1.4	0.70	LSASS access, SAM dump, or similar memory access pattern

Compliance Pack

Audit log integrity, policy violation clustering, and documentation signals for SOC 2 / ISO 27001 / CMMC environments

Phase 2

Detector	Signal Type	Weight	Gate	Notes
Policy Violation Clustering	Compliance	1.2	0.60	Multiple policy events from same identity or system in compressed window
Audit Log Tampering Signal	Integrity	1.4	0.75	Log deletion, modification, or gap — integrity breach of evidence chain
Data Retention Boundary Violation	Data Governance	1.2	0.60	Data accessed or modified beyond its defined retention policy scope
Regulatory Access Anomaly	Access Anomaly	1.2	0.60	Access to regulated data classes by identities without documented authorization
Configuration Change Outside Window	Change Mgmt	1.1	none	System configuration changed outside approved change management window
Segregation of Duties Violation	Governance	1.3	0.65	Single identity performing actions that require dual authorization

OT / Industrial Pack

SCADA, ICS, and critical infrastructure behavioral signals — where a false negative is not an alert, it's a consequence

Phase 3

OT environments have one defining characteristic that makes AFI's architecture uniquely appropriate: the baseline for 'normal' is extremely stable. A pump that runs at 60Hz, a valve that opens on a 4-hour cycle, a temperature sensor that varies within a 2-degree band — these are among the most consistent behavioral baselines in any sector. Deviation from them is meaningful by definition.

Detector	Signal Type	Weight	Gate	Notes
SCADA Access Anomaly	Access / Control	1.4	0.70	Unusual identity or timing accessing SCADA supervisory layer
OT/IT Boundary Crossing	Lateral Movement	1.4	0.75	IT-side identity or process initiating connections into OT network segment
Industrial Control Access Anomaly	Access / Control	1.3	0.65	PLC or DCS access outside maintenance windows or authorized operator list
Setpoint Modification Pattern	Integrity	1.4	0.75	Process setpoint changes deviating from engineering baseline — Stuxnet class
HMI Access from Network Anomaly	Access Anomaly	1.3	0.65	HMI accessed from system or IP not in approved operator list
Engineering Workstation Anomaly	Endpoint / OT	1.3	0.65	Unusual file access or network traffic on OT engineering workstations
Firmware / Logic Download Attempt	Integrity	1.4	0.75	Attempt to write firmware or logic to PLC outside maintenance window
Process Telemetry Suppression	Evasion	1.3	0.70	Reduction or interruption in

Detector	Signal Type	Weight	Gate	Notes
				expected OT telemetry stream — visibility attack

Insurance Underwriting Pack

Risk posture signals for cyber insurance carriers — AFI as the evidence substrate for policy issuance, renewal, and breach investigation

Phase 3

This pack operates on two levels simultaneously. First: detecting anomalies that indicate elevated insurance risk in a covered environment. Second: generating the audit evidence that carriers require for underwriting decisions. AFI doesn't just protect insured environments — it becomes part of the policy issuance workflow.

Detector	Signal Type	Weight	Gate	Notes
MFA Coverage Gap Detection	Access Control	1.2	none	Identities authenticating without MFA to sensitive systems — underwriting risk signal
Privileged Account Hygiene Anomaly	Privilege	1.2	none	Stale admin accounts, excessive standing privilege — policy risk indicators
Backup Access Anomaly	Integrity	1.2	0.60	Unusual access to backup systems — ransomware precursor and insurance red flag
Recovery System Integrity Signal	Integrity	1.2	0.60	Backup deletion, modification, or disconnection from production systems
Data Classification Access Mismatch	Data Access	1.1	none	Sensitive data class accessed by identity without documented authorization
Repeat Incident Pattern	Behavioral	1.2	0.60	Same behavioral signature recurring — indicates unresolved systemic weakness

Detector	Signal Type	Weight	Gate	Notes
Incident Response Delay Signal	Governance	1.0	none	Alert fired but no analyst action within policy-defined response window

Positioning for insurance carriers: AFI is the first product that can serve as both a prevention tool and an audit evidence substrate simultaneously. An insurer that requires AFI deployment as a policy condition has materially reduced their exposure and gained a real-time risk signal on every covered environment.

Static Systems / Physical Security Pack

The 'is your door open' baseline — physical-digital convergence signals, badge anomalies, and camera pattern analysis without semantic content

Phase 3

You identified this correctly: the static systems category is deceptively simple and deceptively powerful. A door access control system has a behavioral baseline. The same employee enters through the same entrance between 8:47am and 9:12am every weekday. When they enter at 2:34am on a Sunday, the door doesn't know it's wrong. AFI does. The detection is behavioral, not semantic — no video content is ever parsed.

Detector	Signal Type	Weight	Gate	Notes
After-Hours Physical Access Anomaly	Physical / Behavioral	1.1	none	Badge access to facility or sensitive area outside employee's historical pattern
Tailgating Pattern Signal	Physical / Behavioral	1.2	0.60	Multiple badge reads in rapid succession — tailgating behavioral signature
Restricted Area Access Sequence	Access Control	1.2	0.60	Access to restricted zones by identity without documented authorization
Physical + Digital Access Correlation	Meta-Signal	1.3	0.65	Physical badge access correlated with simultaneous unusual digital activity
Camera Zone Entry / Exit Anomaly	Behavioral	1.0	none	Motion pattern in monitored zone deviating from baseline occupancy profile
Access Credential Sharing Signal	Identity	1.2	0.60	Same physical credential used in physically impossible sequence (2 buildings, 3 min)

Critical architecture note: this pack uses structural behavioral signals only — timing, sequence, identity correlation. No video content is ever parsed or stored. AFI never becomes a surveillance tool. It becomes a behavioral anomaly detector that happens to consume physical access signals. The distinction is the product.

Supply Chain Pack

Vendor access anomalies, third-party credential abuse, and data flow integrity signals across extended enterprise boundaries

Phase 3—4

Detector	Signal Type	Weight	Gate	Notes
Third-Party Access Pattern Anomaly	Access / Vendor	1.2	0.60	Vendor or contractor accessing systems outside contracted scope or hours
Vendor Credential Sharing Signal	Identity	1.2	0.60	Third-party credentials exhibiting behavioral signatures of shared use
Software Component Integrity Signal	SBOM / Integrity	1.3	0.65	Software update or dependency change outside approved SBOM baseline
API Consumer Behavior Anomaly	API / Integration	1.1	none	Third-party API integration deviating from its documented call pattern
Data Flow Boundary Violation	Exfiltration	1.3	0.65	Data crossing supply chain boundary outside approved integration flows
Vendor Privilege Escalation	Privilege	1.3	0.65	Third-party identity acquiring permissions beyond contracted role baseline
Update Pipeline Integrity Signal	Integrity	1.3	0.70	Software update delivery mechanism exhibiting signs of tampering or redirection

Telecommunications Pack

Coordinated call fraud, robocall patterns, and network anomalies — the Verizon use case, applied to any carrier environment

Phase 4

Telecom fraud has behavioral signatures. A robocall campaign doesn't use random calling patterns — it has rate characteristics, targeting sequences, and geographic spread signatures. A compromised PBX routes calls through predictable relay sequences. These are exactly the behavioral geometries AFI's fusion engine was built to detect.

Detector	Signal Type	Weight	Gate	Notes
Coordinated Call Pattern Detection	Fraud / Behavioral	1.3	0.65	Multiple origination points exhibiting correlated call timing and targeting
Robocall Rate Anomaly	Volume / Behavioral	1.3	0.65	Call initiation rate exceeding human-possible baseline from single origin
PBX / SIP Trunk Anomaly	Infrastructure	1.2	0.60	Unusual routing, toll fraud indicators, or unauthorized trunk usage
International Call Pattern Deviation	Fraud	1.2	0.60	International call destination pattern deviating from subscriber baseline
STIR/SHAKEN Attestation Anomaly	Identity	1.1	none	Call attestation level inconsistent with origination profile
Subscriber Behavior Anomaly	Behavioral	1.1	none	Individual subscriber call pattern deviation from historical baseline
Network Infrastructure Access Anomaly	Access / Infra	1.3	0.65	Unusual access to core network management systems or call routing

AI Agent Security Pack

Phase 4 — High
Priority

The fastest-growing attack surface in the world — agents acting out of scope, prompt injection at scale, and agentic privilege abuse

AI agents do not make AFI obsolete — they make it more valuable. An AI agent scanning for threats faces the same alert fatigue problem as a human analyst. Feed it 42,000 raw alerts and it is an expensive version of the same bottleneck. AFI solves that upstream. The fusion engine produces a structured incident object that AI agents consume. AFI becomes the signal layer that agentic security systems require.

Detector	Signal Type	Weight	Gate	Notes
Agent Out-of-Scope Action	Agentic / Behavioral	1.3	0.65	AI agent accessing resources, APIs, or systems outside its defined operational scope
Prompt Injection Behavioral Signal	Agentic / Integrity	1.3	0.65	Agent behavioral deviation consistent with instruction override — prompt injection
Agentic Privilege Escalation	Privilege	1.3	0.65	Agent acquiring permissions or capabilities beyond its defined authorization
Agentic Data Exfiltration Pattern	Exfiltration	1.4	0.70	Agent transferring data to external endpoints outside workflow definition
Agent-to-Agent Communication Anomaly	Agentic / Network	1.2	0.60	Unexpected agent communication patterns — lateral movement in agentic layer
Tool Use Anomaly	Agentic / Behavioral	1.2	0.60	Agent invoking tools in sequences or frequencies outside operational baseline
Agent Identity Spoofing Signal	Identity	1.3	0.65	Agent presenting identity claims inconsistent with its provisioned credentials
Agentic Loop / Runaway Detection	Operational	1.1	none	Agent executing repetitive action sequences beyond defined task scope

Competitive note: no current security vendor has a mature agentic security offering. The attack surface is 18 months ahead of the defensive tooling. AFI's behavioral architecture — specifically designed for signal fusion without semantic content parsing — is architecturally correct for this problem. Agents produce rich behavioral signals. AFI reads behavioral signals. The fit is structural, not forced.

Robotics & Autonomous Systems Pack

The behavioral baseline for a robot doing its job versus a robot that has been compromised — humanoid, industrial, and logistics systems

**Phase 4–5 —
Strategic**

100,000 humanoid robots is not science fiction. Tesla Optimus, Figure AI, Boston Dynamics, and others are deploying at scale. Each robot produces rich telemetry: joint torque, velocity, positioning, task sequencing, communications activity, and environmental sensor readings. Normal operation has a behavioral signature. Compromise, misconfiguration, or adversarial injection has a different one. AFI was built to detect the difference.

Detector	Signal Type	Weight	Gate	Notes
Robot Motion Pattern Anomaly	Behavioral / OT	1.2	none	Movement sequence deviation from task baseline — path, force, timing
Robotic Firmware / Logic Tamper Signal	Integrity	1.4	0.75	Firmware update or logic modification outside approved maintenance window
Fleet Coordination Anomaly	Coordinated / Behavioral	1.2	0.60	Unusual synchronization or communication patterns across robot fleet
Sensor Data Integrity Signal	Integrity	1.2	0.60	Sensor readings deviating from environmental baseline — possible spoofing
Command Override Attempt	Integrity / Control	1.4	0.75	Control command received from source outside authorized operator list
Robot-to-Infrastructure Access Anomaly	Lateral Movement	1.3	0.65	Robot system initiating unexpected connections to enterprise infrastructure
Behavioral Drift Post-Update	Integrity / Behavioral	1.2	0.60	Post-firmware robot behavioral profile diverging from pre-update baseline

Long-horizon positioning: AFI doesn't need to be the robotics security brand. AFI needs to be the behavioral intelligence layer that robotics security platforms build on top of. The architecture is already there. The Phoenix Signal Network compounds across robotic deployments exactly as it does across enterprise security deployments. History compounds. A new vendor cannot replicate it.

Defense & Intelligence Pack

Command, targeting, and decision architecture — the software intelligence layer above hardware primes, not competing with them

**Phase 5 —
Strategic**

The defense market is not reached directly from an MSP starting position. It is reached through demonstrated capability in civilian environments, CMMC compliance credentialing, and a track record of accurate behavioral detection at scale. The Phoenix Signal Network's cross-environment intelligence — patterns seen across thousands of deployments before any individual operator sees them — is exactly what creates strategic urgency for a defense acquirer. Not today. But the architecture leads there.

Detector	Signal Type	Weight	Gate	Notes
Command Chain Integrity Signal	Integrity / Auth	1.4	0.75	Authentication or authorization anomaly in command-and-control infrastructure
Cross-Classification Boundary Signal	Data Access	1.4	0.75	Data access pattern suggesting classification boundary violation
Supply Chain Adversarial Signal	Supply Chain	1.3	0.70	Software or hardware component behavioral signals consistent with supply chain compromise
Insider Threat Behavioral Pattern	Insider	1.3	0.65	Behavioral sequence consistent with insider threat — data accumulation, unusual access
SCIF / Controlled Environment Anomaly	Physical / Digital	1.3	0.70	Physical or digital access anomaly in controlled access areas
Nation-State TTP Signature	Threat Intel	1.4	0.75	Behavioral patterns consistent

Detector	Signal Type	Weight	Gate	Notes
				with known nation-state actor TTPs via Phoenix Network
Weapons System Integration Anomaly	OT / Integrity	1.4	0.80	Behavioral deviation in integrated weapons or platform management systems
Communication Security Anomaly	Comms / Integrity	1.3	0.65	Encrypted communication pattern anomaly — timing, volume, or routing deviation

The Expansion Architecture

Every detector pack listed in this document is built on the same foundation: the fusion engine evaluates behavioral geometry, not content. Adding a new sector requires new connectors and new baseline definitions. The fusion math doesn't change. The Phoenix Signal Network benefits from every new deployment regardless of sector.

What Each New Pack Actually Requires

Component	What It Requires	Shared Across Packs?
Connector	New data source adapter translating sector signals into normalized event schema	No — built once per source
Detector Pack	New behavioral rules with weights and gates appropriate to sector	No — sector-specific
Baseline Definitions	What 'normal' looks like in this sector — differs by environment	No — environment-calibrated
Fusion Engine	Unchanged. Receives normalized findings. Produces severity scores.	Yes — universal
Phoenix Signal Network	Unchanged. Receives abstracted signatures. Compounds across all deployments.	Yes — cross-sector
Incident Intelligence Report	AI annotation layer explains findings. Prompt library expands per sector.	Mostly — sector context added

The Phoenix Signal Network — Cross-Sector Compounding

The Phoenix Signal Network operates on abstracted behavioral signatures — never raw events, never semantic content. When the network sees a behavioral pattern in a healthcare environment that shares geometric characteristics with a financial fraud precursor, that correlation is available to every deployment in the network. No individual deployment can develop this cross-sector pattern recognition independently. The network sees it because the network spans sectors.

This is the moat that cannot be replicated from a standing start. A new entrant can build a fusion engine. They cannot build 18 months of cross-sector, cross-environment behavioral signal history. The Phoenix Signal Network makes history itself the competitive advantage.

The Sequence That Makes Sense

Phase	Timeline	Focus	What It Unlocks
1	Now – 2 years	MSPs, M365, MSP Operations Pack, Healthcare, Financial	Case studies, revenue, Phoenix Network seed data, IT Nation presence

Phase	Timeline	Focus	What It Unlocks
2	2–4 years	Azure/AWS Marketplace, Cloud Pack, Email, Endpoint, Compliance	Enterprise direct sales, platform ecosystem credentialing
3	3–6 years	OT/Industrial, Insurance, Supply Chain, Static Systems	Sector depth, regulated market penetration, carrier partnerships
4	5–8 years	Telecom, AI Agent Security, Robotics	Adjacent market expansion, agentic security leadership position
5	7–10 years	Defense & Intelligence	Strategic acquirer positioning, government contract eligibility

The engine is built. The architecture is correct.

Every sector in this document follows from first principles — not analogy, not optimism.

The ceiling will reveal itself. Build the foundation right.